



Fig. 17: Under-sea warfare helicopter



Fig. 18: A sailor loads dispensable sonobuoys in an under-sea helicopter (left); an under-sea warfare helicopter carrying a torpedo (right) and dispensable sonobuoys can be seen above it

of destroyer. This system creates a picture of the underwater scenario by comparing the received echoes.

Each sonobuoy receives an echo. Depending upon their locations, the echo is received late or early. Through a process called triangulation, exact location of the submarine can be pinned down. Not just one helicopter, but helicopters from other destroyers are pressed into service, depending upon the combat situation. Because once a submarine's presence is detected, the CSG puts in every helicopter to detect and neutralise that submarine. A fairly clear underwater picture can be evolved by these destroyers.

If the submarine is detected within the range of a helicopter, the under-sea warfare C² system from the destroyer sends launch commands for a helicopter to launch a helicopter-launched torpedo.

The carrier, which is the prize target for a submarine, ironically, has neither strike capability against submarines nor strong organic submarine-detection capability. It has to defend against the submarine only when the submarine passes through the perimeter created by destroyers.

The only defence it has against enemy submarines is through a torpedo decoy to seduce away the torpedoes launched by enemy submarines.

Torpedoes sense the noises *chug chugs* and *murmurs* generated by the target's propellers and engine, respectively. There are certain torpedoes that track the wake created by the ships. The torpedo decoy is towed behind the carrier at a long distance. This decoy has a sonar that only looks for the sound of an approaching torpedo. When it detects an approaching torpedo, it produces a sound mimicking the sound produced by the carrier's propellers and engine murmurs. The sound is so powerful and realistic that the torpedo is clearly attracted towards the decoy instead of the ship and thus gets spoofed. Employment of this decoy is a worst-case scenario in which associating ships have got sunk by enemy action.

Surface warfare

If a large flotilla of enemy surface ships is spotted faraway by E-2C airborne early warning (AEW) aircraft, F-18 combat aircraft carrying anti-ship missiles may be sent to fin-

ish these off. This is the long-range engagement but it is the destroyers that normally take care of medium-range engagements. Aegis C² system keeps detecting and tracking surface targets. This is done by AN/SPS-67 surface-search radar, AN/SPY-1 radar and AN/SPG-9 radar.

For medium-range engagements, anti-ship missiles called Harpoons are used. These missiles are capable of reaching 100km. For short-range engagements, 12.6cm (5-inch) guns, or, simply, guns, are used to engage in a classical way.

In short-range engagements, the range, bearing and speed of the target are continuously updated and transferred to the gun's fire control system (FCS). The FCS plots the course of the enemy surface vessel. Then, it calculates the elevation and orientation of the gun to be maintained and actuates the gun, accordingly. It even takes wave motion into account. If the destroyer is heaved one metre up, the shell may land far away from the intended target. So the FCS stabilises the gun accordingly and maintains it in the required orientation.

At the command of the gunner, the FCS fires the gun. This gun can fire in rapid succession since the shells are loaded into the gun through an automatic loader. The destroyer has one such gun and the cruiser has two (one at the front and another at the back).

For medium-range engagements, Aegis system feeds the target coordinates to Harpoon anti-ship missile's FCS. For targets beyond the radar horizon, the carrier-borne AEW aircraft or even satellites provide the target's location. These transmit data to the carrier from where it is transferred to the Aegis C² system and then to Harpoon's FCS.

Harpoon's FCS derives the trajectory to be followed by the missile and programmes it into the navigation system of the missile. When fired, Harpoon being a fire-and-forget missile flies itself towards the target and destroys it. Initially, Harpoon uses its